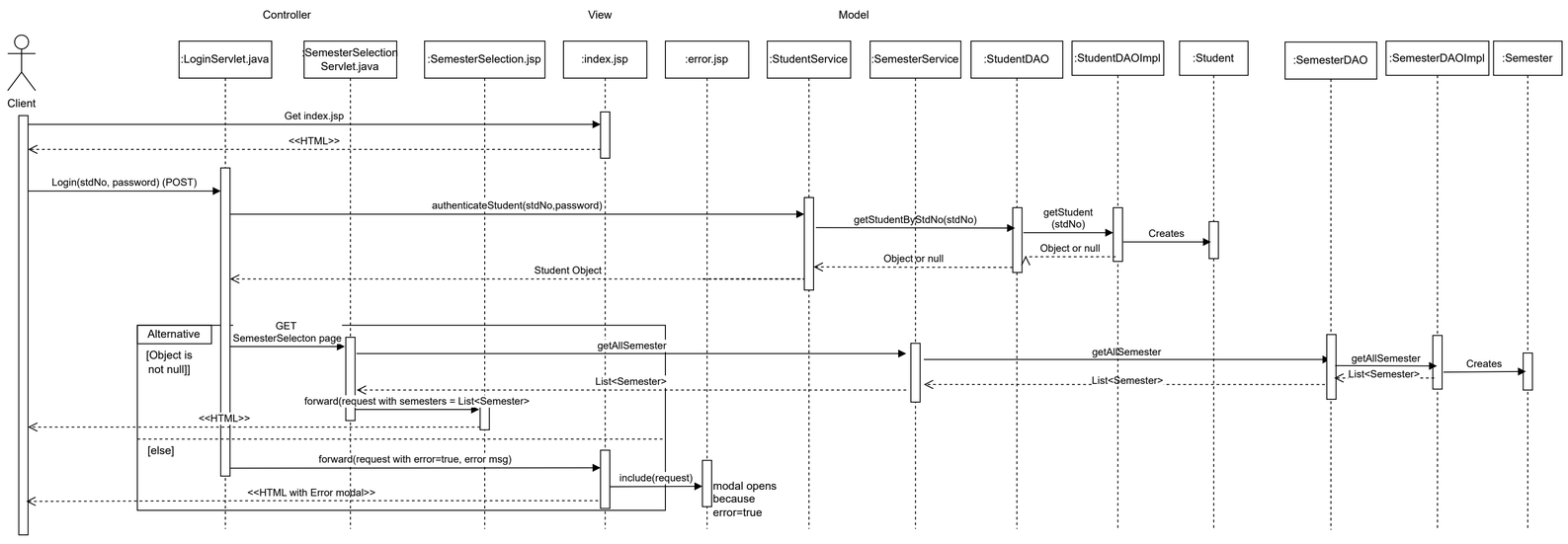
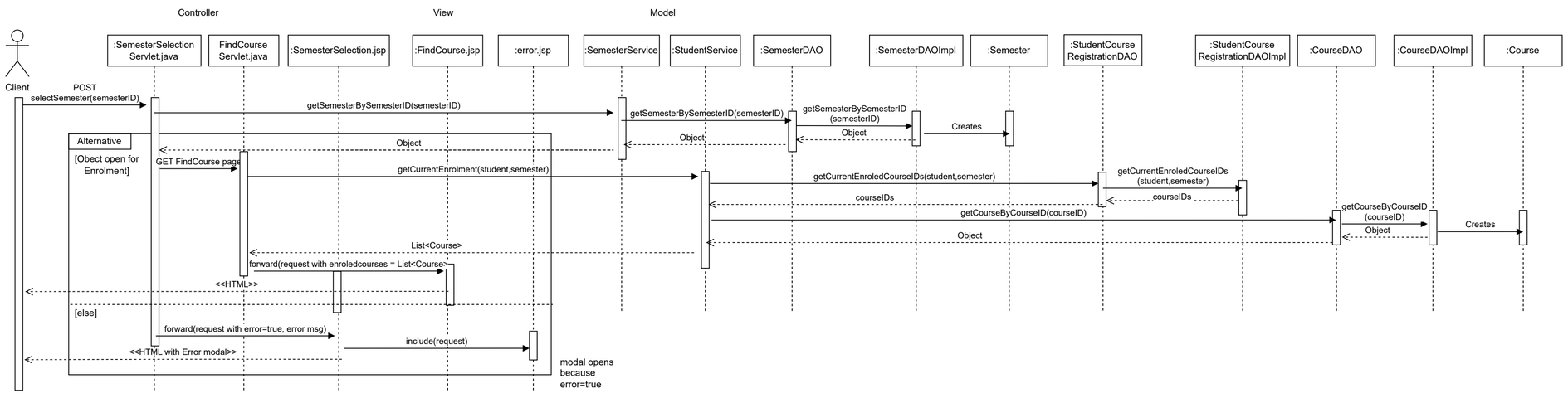
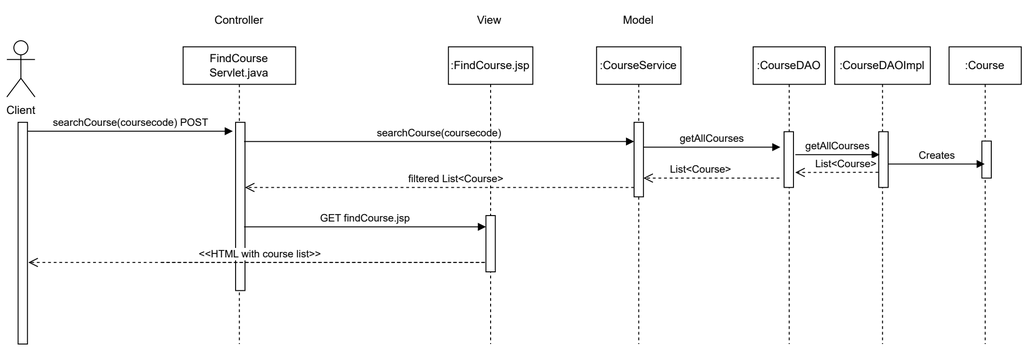
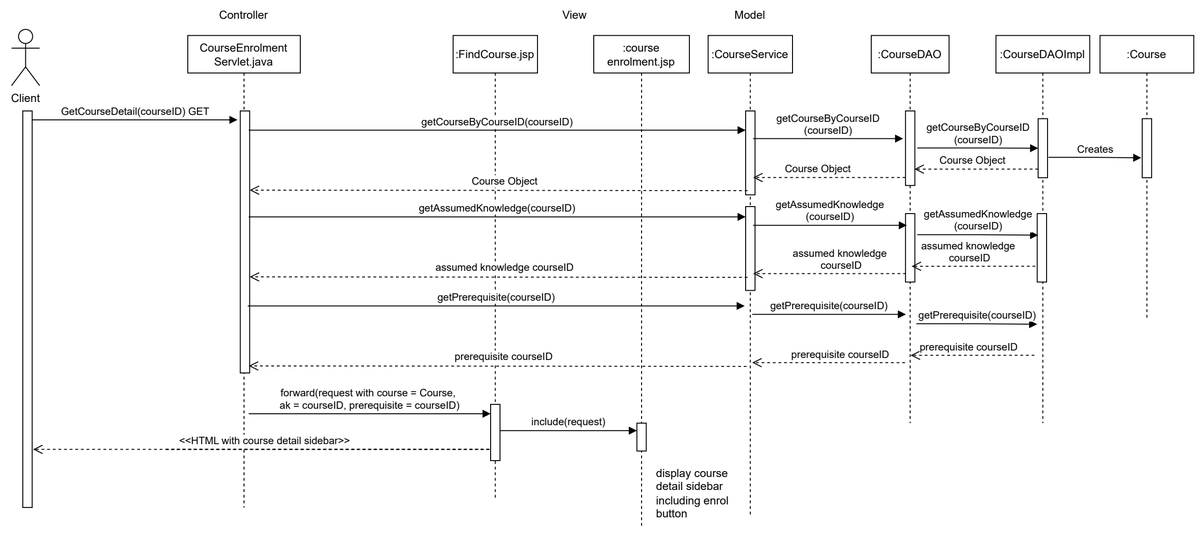
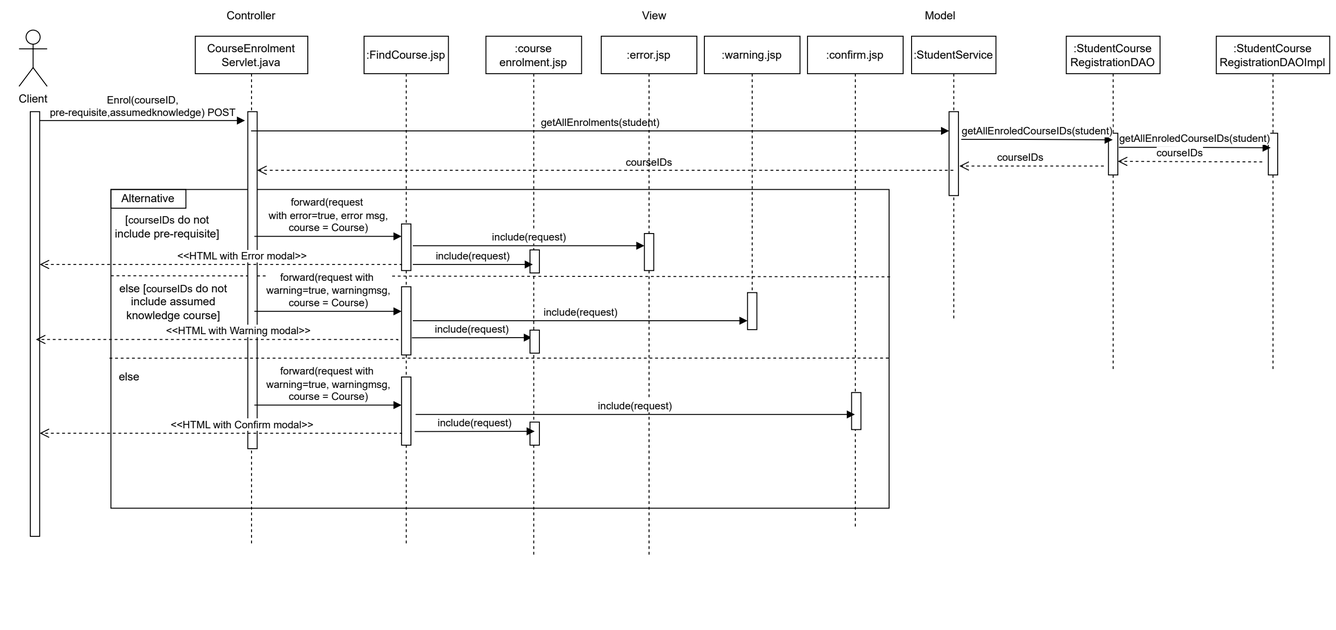
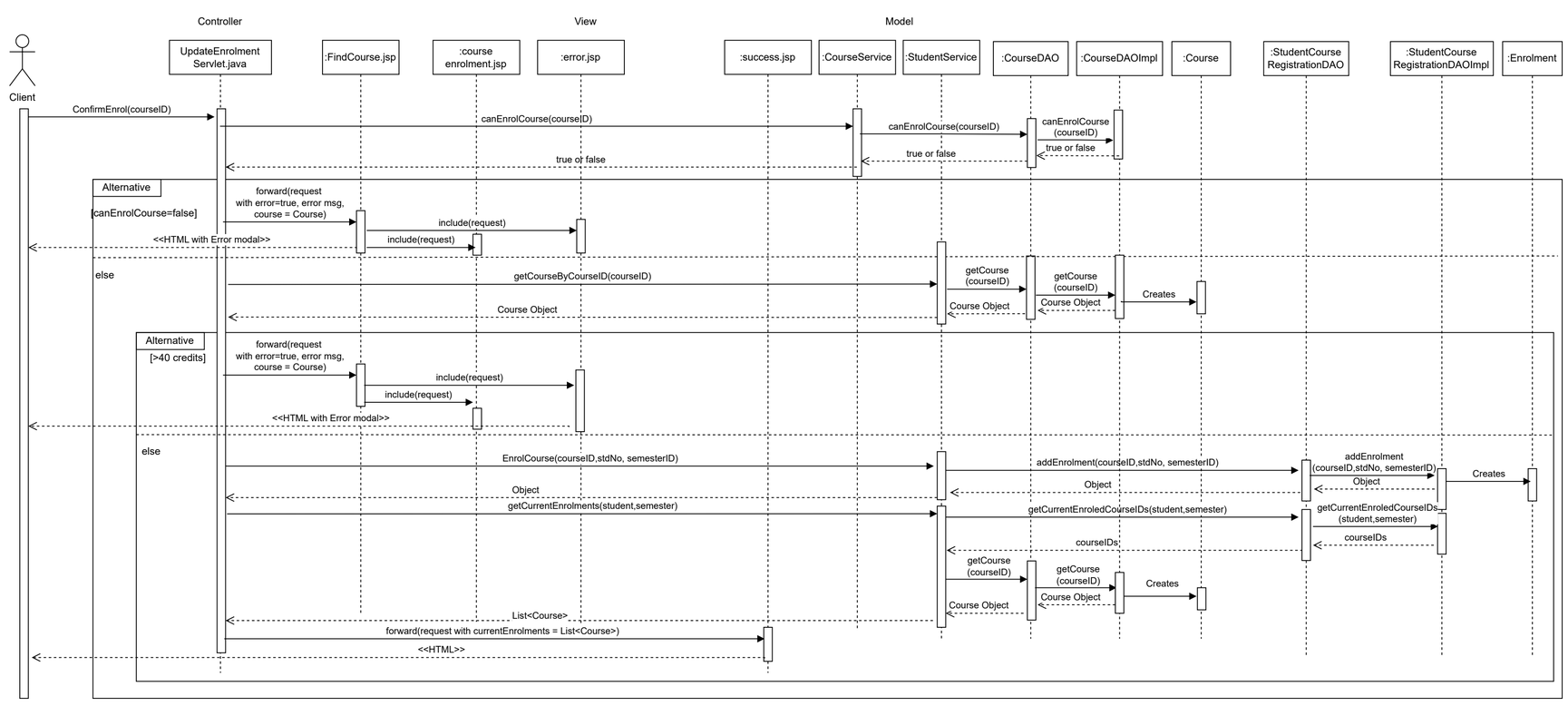

Controller
View
Model
Client
:LoginServlet.java
:SemesterSelection Servlet.java
:SemesterSelection.jsp
:index.jsp
:error.jsp
:StudentService
:SemesterService
:StudentDAO
:StudentDAOImpl
:Student
:SemesterDAO
:SemesterDAOImpl
:Semester
Get index.jsp
<<HTML>>
Login(stdNo, password) (POST)
authenticateStudent(stdNo,password)
getStudentByStdNo(stdNo)
getStudent (stdNo)
Creates
Object or null
Object or null
Student Object
Alternative
[Object is not null]]
GET SemesterSelecton page
getAllSemester
getAllSemester
getAllSemester
Creates
List<Semester>
List<Semester>
List<Semester>
forward(request with semesters = List<Semester>
<<HTML>>
[else]
forward(request with error=true, error msg)
include(request)
modal opens because error=true
<<HTML with Error modal>>


Controller
View
Model
Client
:SemesterSelection Servlet.java
FindCourse Servlet.java
:SemesterSelection.jsp
:FindCourse.jsp
:error.jsp
:SemesterService
:StudentService
:SemesterDAO
:SemesterDAOImpl
:Semester
:StudentCourse RegistrationDAO
:StudentCourse RegistrationDAOImpl
:CourseDAO
:CourseDAOImpl
:Course
POST selectSemester(semesterID)
getSemesterBySemesterID(semesterID)
getSemesterBySemesterID(semesterID)
getSemesterBySemesterID (semesterID)
Creates
Object
Object
Object
Alternative
[Obect open for Enrolment]
GET FindCourse page
getCurrentEnrolment(student,semester)
getCurrentEnroledCourseIDs(student,semester)
getCurrentEnroledCourseIDs (student,semester)
courseIDs
courseIDs
getCourseByCourseID(courseID)
getCourseByCourseID (courseID)
Object
Creates
Object
List<Course>
forward(request with enroledcourses = List<Course>
<<HTML>>
[else]
forward(request with error=true, error msg)
include(request)
modal opens because error=true
<<HTML with Error modal>>


Controller
View
Model
Client
FindCourse Servlet.java
:FindCourse.jsp
:CourseService
:CourseDAO
:CourseDAOImpl
:Course
searchCourse(coursecode) POST
searchCourse(coursecode)
getAllCourses
getAllCourses
Creates
List<Course>
List<Course>
filtered List<Course>
GET findCourse.jsp
<<HTML with course list>>


Controller
View
Model
Client
CourseEnrolment Servlet.java
:FindCourse.jsp
:course enrolment.jsp
:CourseService
:CourseDAO
:CourseDAOImpl
:Course
GetCourseDetail(courseID) GET
getCourseByCourseID(courseID)
getCourseByCourseID (courseID)
getCourseByCourseID (courseID)
Creates
Course Object
Course Object
Course Object
getAssumedKnowledge(courseID)
getAssumedKnowledge (courseID)
getAssumedKnowledge (courseID)
assumed knowledge courseID
assumed knowledge courseID
assumed knowledge courseID
getPrerequisite(courseID)
getPrerequisite(courseID)
getPrerequisite(courseID)
prerequisite courseID
prerequisite courseID
prerequisite courseID
forward(request with course = Course, ak = courseID, prerequisite = courseID)
include(request)
<<HTML with course detail sidebar>>
display course detail sidebar including enrol button


Controller
View
Model
Client
CourseEnrolment Servlet.java
:FindCourse.jsp
:course enrolment.jsp
:error.jsp
:warning.jsp
:confirm.jsp
:StudentService
:StudentCourse RegistrationDAO
:StudentCourse RegistrationDAOImpl
Enrol(courseID, pre-requisite,assumedknowledge) POST
getAllEnrolments(student)
getAllEnroledCourseIDs(student)
getAllEnroledCourseIDs(student)
courseIDs
courseIDs
Alternative
[courseIDs do not include pre-requisite]
forward(request with error=true, error msg, course = Course)
include(request)
include(request)
<<HTML with Error modal>>
else [courseIDs do not include assumed knowledge course]
forward(request with warning=true, warningmsg, course = Course)
include(request)
include(request)
<<HTML with Warning modal>>
else
forward(request with warning=true, warningmsg, course = Course)
include(request)
include(request)
<<HTML with Confirm modal>>


Controller
View
Model
Client
UpdateEnrolment Servlet.java
:FindCourse.jsp
:course enrolment.jsp
:error.jsp
:success.jsp
:CourseService
:StudentService
:CourseDAO
:CourseDAOImpl
:Course
:StudentCourse RegistrationDAO
:StudentCourse RegistrationDAOImpl
:Enrolment
ConfirmEnrol(courseID)
canEnrolCourse(courseID)
canEnrolCourse(courseID)
canEnrolCourse (courseID)
true or false
true or false
true or false
Alternative
[canEnrolCourse=false]
forward(request with error=true, error msg, course = Course)
include(request)
include(request)
<<HTML with Error modal>>
else
getCourseByCourseID(courseID)
getCourse (courseID)
getCourse (courseID)
Creates
Course Object
Course Object
Course Object
Alternative
[>40 credits]
forward(request with error=true, error msg, course = Course)
include(request)
include(request)
<<HTML with Error modal>>
else
EnrolCourse(courseID,stdNo, semesterID)
addEnrolment(courseID,stdNo, semesterID)
addEnrolment (courseID,stdNo, semesterID)
Creates
Object
Object
getCurrentEnrolments(student,semester)
getCurrentEnroledCourseIDs(student,semester)
getCurrentEnroledCourseIDs (student,semester)
courseIDs
courseIDs
getCourse (courseID)
getCourse (courseID)
Creates
Course Object
Course Object
List<Course>
forward(request with currentEnrolments = List<Course>)
<<HTML>>